



A joint possibilistic data association technique for tracking multiple targets in a cluttered environment



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ABSTRACT

Multitarget tracking in a cluttered environment is a significant issue with a wide variety of applications. A typical approach to address this issue is the joint probabilistic data association (JPDA) technique. This technique determines joint probabilities over all targets and hits and updates the predicted target state estimate using a probability-weighted sum of innovations. This paper proposes a new joint possibilistic data association technique for tracking multiple targets. Unlike the JPDA technique, the proposed technique determines joint possibilities over all targets and hits and updates the predicted target state estimate using a possibility-weighted sum of innovations. The possibility weights are determined using the noise covariance matrices and the current received measurements such that the total sum of the distances between all measurements and targets is minimized. The proposed technique performs data association based on a possibility matrix of measurements to trajectories; thus, it highly reduces the computational complexity compared to conventional data association techniques. The proposed association technique is applied to examples of multitarget tracking in a cluttered environment, and the results demonstrate its efficiency.

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1. Introduction

Data fusion systems have important civilian and military applications [2,6,8–10,22,29,32,37,48,57,58]. One of the main issues with data fusion systems is multiple-target tracking [1,11,35,38,39,55]. Data association is an important issue when tracking multiple targets in the presence of false alarms or clutter [3–5,7,52,56]. Data association is applied in many practical systems such as wireless sensor networks [33,36,46], vehicle tracking [19], data mining [23,26,41], face recognition [31], meteorological data [17], dynamic images [60], and others.

Multiple-target tracking techniques include trajectory initiation techniques, trajectory maintenance techniques, and trajectory deletion techniques. We focus only on data association, which is the main task of trajectory maintenance techniques. There are other techniques in the literature to solve the problems of trajectory initiation and trajectory deletion [12–16].

Deciding from which target, if any, a received hit (measurement) originated is one of the most crucial issue in multitarget tracking systems. This is referred to as the data association process. Data association is a fundamental process in multitarget tracking systems. It is the process of deciding which of the received hits to use to update each target trajectory (track), i.e. it is responsible for partitioning the received hits into groups that could have originated from the same target trajectories. When tracking multiple targets, there is always uncertainty associated with the origin of the received hits, when there is

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a high false alarm, a low detection probability, a cluttered environment, or multiple targets in the same neighborhood. Hits can be in a wide variety of forms, such as x - and y -positions, range, bearing, etc. After performing data association, the estimate of the state for each target trajectory is updated with the associated target hits. Data association and state estimation have been the subjects of extensive study since the 1970s [13–17]. This is called hits-to-targets association for dynamic targets. In most real multitarget tracking problems, association of hits is performed solely on the basis of positional information.

There are two main categories of data association in multiple targets tracking: nearest-neighbor data association and all-neighbor data association. The nearest-neighbor data association category has two types of solutions: strongest neighbor filter (SNF) and nearest-neighbor filter (NNF) [12–16]. In these approaches, one hit, at most, can be used to update a given target trajectory. In the SNF, the signal with the highest intensity among the validated received hits is selected to update the target trajectory. In the NNF, the hit that is closest to the predicted target hit is the one selected. In nearest-neighbor data association approaches hits are assigned to existing trajectories in such a way as to minimize or maximize an overall correlation measure. A clear description of nearest-neighbor association techniques is provided in [16]. The performance of the nearest-neighbor association technique under many correlation measures is considered in many studies [28,53]. The optimum nearest-neighbor data association filter is presented in [12–14] by maximizing the likelihood of the residual error. The SNF and NNF techniques perform very well in the case of well-spaced targets with high signal-to-noise ratios and a clutter-free environment. However, they do not perform well in the case of closing targets with high false alarm rates or low detection probabilities [1,7,12,16].

The optimal solution of the multitarget association problem is the multiple hypothesis tracker (MHT) [12,16]. This is considered to be the theoretically optimum data association approach under idealized assumptions. The MHT addresses the multitarget association problem by evaluating the likelihood that there is a target given a number of hits. The output from the MHT is typically a list of hypotheses that can be arranged by their probability estimates. Unfortunately, the high computational complexity of MHT limits its practical implementation. In addition, a priori knowledge of the noise statistics is required. Therefore, suboptimal realization of the MHT has many potential pitfalls. For this reason, many tracking approaches have been developed which sacrifice performance for the sake of computational feasibility. Thus, the newly developed approaches are suboptimal but computationally feasible. The probabilistic data association (PDA) filter and the joint probabilistic data association (JPDA) filter were developed in [12] to solve the problem of data association in the case of multiple-target tracking. Instead of using only one hit among all the received and validated hits while discarding the others, the PDA and JPDA filters use all of the validated hits with different probability weights. The result is that the updated target trajectory may contain contributions from more than one hit, each with some correlation probability. The JPDA filter is identical to the PDA filter except that the correlation probabilities are computed using all hits and all trajectories. An approximation of the JPDA filter is proposed in [25] to reduce the required computations. This approximation highly degrades the performance in the case of a large number of hits and targets [61,62].

While the MHT is classified as a hard-decision multiscan association technique, the interacting multiple model joint probabilistic data association (IMMJPDA) is classified as a soft-decision zero-back scan association technique [7,47]. The IMMJPDA combines several trajectories in a probabilistic way to update each target state estimate [12,40]. This is one of the best association approaches, yielding a performance comparable to that of the MHT approach. There are other schemes for tracking multiple targets based on expectation maximization [13,14], the probability hypothesis density (PHD) filter [43], the particle filter [30,54], the Kalman-Levy filter [51], and the inverse gamma distributed texture [7,14]. The time-recursive multiple targets tracking approach, based on expectation maximization, achieves better performance compared to the JPDA technique at the expense of additional computations [1,7,15].

Neural networks and fuzzy logic approaches are considered approximate solutions to the problem of multiple targets tracking [18,20,24,27,45]. The disadvantages of neural network implementations are that they require a very large number of neurons and require training with an unreasonable number of trajectories. For example, a neural network scheme is presented in [36] to implement the JPDA approach. Due to the complexity of this scheme, it can handle only a small number of targets and hits (6 targets and 20 hits at most). In general, the number of neurons grows exponentially with the number of targets and the number of hits [1,34,49].

Fuzzy logic technologies are also used in multitarget tracking [5–7,42,44]. Fuzzy logic approaches for data association consist of four basic elements [1,3,4,7]: (1) fuzzification interface, which transforms the received sensor hits into fuzzy (linguistic) variables using some forms of membership functions; (2) fuzzy knowledge-base, which contains fuzzy logical statements in terms of fuzzy IF-THEN rules. The IF-THEN rules use linguistic terms (such as low, medium, and high) to incorporate ambiguity; (3) fuzzy inference engine, which employs fuzzy knowledge-base to generate fuzzy outputs; and (4) defuzzification interface, which transforms the fuzzy outputs (from the fuzzy inference engine) into non fuzzy outputs to obtain the final association decisions.

The disadvantages of fuzzy logic approaches are that they require a very large number of fuzzy IF-THEN rules and/or use iterative approaches. A fuzzy approach, based on the fuzzy clustering means (FCMs) algorithm, is proposed in [7]. It replaces the probability scores with fuzzy scores. The fuzzy scores are determined using a time-consuming iterative method, and the number of iterations depends on the initial values, the number of targets, and the number of measurements. In general, fuzzy logic solutions of the problem of multiple target tracking are approximate solutions. The accuracy of the solutions depends on the number of input variables, the choice of fuzzy functions, the number of linguistic variables, and the number of fuzzy statements and rules. In addition, the solutions of fuzzy logic approaches for multiple target data association and tracking are

very complicated in practice. For example, the extension of the fuzzy tracking approach developed in [50] in the case of s input variables (such as elevation and azimuth), ℓ linguistic variables (such as small and medium), and n received hits requires $(\ell n)^s$ fuzzy IF-THEN rules [1,7] (assuming that the number of targets is equal to the number of hits). If we assume a multiple-target tracking system with five linguistic variables ($\ell = 5$), three input variables ($s = 3$), nine targets ($n_t = 9$), and nine hits ($n = 9$), the required number of fuzzy IF-THEN rules will be 91125 rules, which is impractical for real time processing. Thus, the extension of fuzzy logic approaches to solve the problem of data association in multiple-target tracking is very complex in the case of large number of targets and hits.

This paper proposes a novel joint possibilistic data association technique for multitarget tracking in a cluttered environment. It incorporates all of the validated hits with corresponding possibility weights to update the target trajectories. It associates hits to trajectories and updates the targets' state estimates using a possibility-weighted sum of innovations. The possibility weights are determined using the covariance matrices and the current sensor hits such that the total sum of the distances between all targets and hits is minimized. Unlike the joint probabilistic data association filter, which determines the joint probabilities over all targets and all validated hits, the proposed technique determines possibility weights between all targets and all validated hits. Unlike fuzzy tracking techniques, which include “fuzzifications”, fuzzy inference mechanisms, and “defuzzifications”, the proposed technique performs data association and multitarget tracking based on a single possibility matrix of hits to targets. Thus, it greatly reduces the required computations compared to other association techniques. The performance of the proposed possibility approach is evaluated and compared to the performance of the JPDA technique and the nearest-neighbor standard filter (NNSF) technique. The results show the efficiency and the simplicity of the proposed association technique.

The remainder of this paper is organized as follows. The problem formulation of multitarget tracking is addressed in Section 2. A brief description of JPDA technique is presented in Section 3. The proposed joint possibilistic data association technique is developed in Section 4. The comparison of computational complexity is analyzed in Section 5. The proposed association technique is applied to some examples of multitarget tracking systems in a cluttered environment in Section 6 to demonstrate its efficiency and feasibility. Performance comparisons with JPDA and NNSF techniques are also presented in Section 6. Finally, Section 7 contains the conclusion.

2. Problem formulation of multitarget tracking

The purpose of this section is to present a general framework and a general concept of multitarget tracking in order to compare the different alternatives with the proposed joint possibilistic one. Multitarget tracking can be viewed as an estimation process but with partial information. If the association of hits to targets is available and performed correctly, it is an easy task to estimate the targets states. In this case, multitarget tracking becomes a traditional estimation problem. So, it is worth noting that the primary difficulty in multitarget tracking lies in data association process because the targets compete for hits.

We assume a multitarget tracking system with n_t targets. Assume that at scan k , n_v number of validated measurements $\{\mathbf{z}_1(k), \mathbf{z}_2(k), \dots, \mathbf{z}_{n_v}(k)\}$ within the target gates are received. The dynamic model of target i is defined as

$$\mathbf{x}_i(k+1) = \mathbf{F}_i \mathbf{x}_i(k) + \mathbf{v}_i(k), \quad i = 1, 2, \dots, n_t, \quad (1)$$

where $\mathbf{x}_i(k)$ is an n -dimensional state vector of target i at scan k , $\mathbf{v}_i(k)$ is a noise input vector, which is called plant noise, and $\mathbf{F}_i$ is an $n \times n$ state transition matrix.

The measurement model of target i is defined as

$$\mathbf{z}_i(k) = \mathbf{H}_i(k) \mathbf{x}_i(k) + \mathbf{w}_i(k), \quad (2)$$

where $\mathbf{z}_i(k)$ is an m -dimensional measurement vector, $\mathbf{w}_i(k)$ is a measurement noise vector, and $\mathbf{H}_i$ is an $m \times n$ measurement matrix. We assume that the measurement noise and the plant noise are uncorrelated, zero mean Gaussian noises with covariance matrices

$$\mathbf{Q}_i(k) = \text{Cov}(\mathbf{v}_i(k)), \quad (3)$$

$$\mathbf{R}_i(k) = \text{Cov}(\mathbf{w}_i(k)). \quad (4)$$

In each scan k , a number of validated hits (n_v) is received. In many radar, sonar, and other tracking systems, hits may originate from the original targets, whose existence and trajectories are generally not known a priori, and also from other random sources, usually termed clutter. In case of multiple targets, the hits may have also originated from one of various targets. Multitarget tracking in this environment initiated trajectories using both target and clutter hits. In this case, the number of hits generated by targets change over time as clutter appear and disappear in the scene. The number of received hits is usually greater than the number of targets ($n_v > n_t$), especially in the case of multitarget tracking in clutter. Target hits are present in each scan k with a certain detection probability (p_d). Given the n_v received hits, Eq. (2) requires determining the track estimates for all existing targets (n_t targets). In the case of closely spaced targets, it is very difficult to determine precisely which target corresponds to each of the received hits. The objective is to determine each target state estimate by using the uncertain hits originating from the targets under surveillance. In a clean (noise-free and clutterless) environment, each target trajectory i is predicted and updated based on its own correct hit using the Kalman filter equations as follows

$$\hat{\mathbf{x}}_i(k+1|k) = \mathbf{F}_i \hat{\mathbf{x}}_i(k|k), \quad (5)$$

$$\mathbf{P}_i(k+1|k) = \mathbf{F}_i \mathbf{P}_i(k) \mathbf{F}_i' + \mathbf{Q}_i(k), \quad (6)$$

$$\hat{\mathbf{x}}_i(k+1|k+1) = \hat{\mathbf{x}}_i(k+1|k) + \mathbf{K}_i(k+1) \tilde{\mathbf{z}}_i(k+1), \quad (7)$$

$$\mathbf{P}_i(k+1|k+1) = [\mathbf{I} - \mathbf{K}_i(k+1) \mathbf{H}_i(k+1)] \mathbf{P}_i(k+1|k), \quad (8)$$

where $\mathbf{K}_i(k)$ is the Kalman filter gain given by

$$\mathbf{K}_i(k+1) = \mathbf{P}_i(k+1|k) \mathbf{H}_i'(k+1) [\mathbf{H}_i(k+1) \mathbf{P}_i(k+1|k) \mathbf{H}_i'(k+1) + \mathbf{R}_i(k+1)]^{-1}, \quad (9)$$

and $\tilde{\mathbf{z}}_i(k+1)$ is the innovation given by

$$\tilde{\mathbf{z}}_i(k+1) = \mathbf{z}_i(k+1) - \mathbf{H}_i(k+1) \hat{\mathbf{x}}_i(k+1|k). \quad (10)$$

The covariance matrix of the innovation (residual error) is given by

$$\mathbf{S}_i(k+1) = \mathbf{H}_i(k+1) \mathbf{P}_i(k+1|k) \mathbf{H}_i'(k+1) + \mathbf{R}_i(k+1). \quad (11)$$

There are two main techniques to solve the problem of associating hits to trajectories in multitarget tracking systems: the multiple hypothesis tracking (MHT) technique and the JPDA technique. The MHT technique utilizes several scans of measurements to associate hits to trajectories. This technique forms multiple hypotheses for trajectory extension but delays making a hard temporal association until more hits from later scans make the association easier. When more hits are available, the highly improbable hypotheses are ignored (or pruned) leaving the trajectories of the actual targets. The increasing number of hypothesis trajectories in MHT increases both the computational complexity and the required memory space. Although track pruning reduces the computational complexity, it yields suboptimal forms of tracking splitting. Unlike the MHT technique, which is a hard data association decision, the JPDA technique is a soft data association decision. The JPDA technique accounts for the conflicting situations of two or more targets in the same extension gate. This technique updates a target trajectory with a weighted sum of feasible measurements in each scan k . The weights are the probabilities that the received hits correctly extend the trajectory. In the JPDA technique, joint probabilities are calculated over all targets and hits. Each joint probability represents the probability of all the individual events combined as a joint event. Thus, the JPDA technique includes the possibility of multiple measurements in one trajectory extension gate coming from other targets. The JPDA technique is well-suited to a high false density environment and can be viewed as an MHT technique which merges hypotheses into weighted mean trajectories on a per scan basis. The JPDA technique requires an exponentially increasing amount of processing time in calculating all the joint probabilities as the number of targets and hits increase. To overcome this difficulty, many suboptimal JPDA approaches have been proposed in the literature. Most of these approaches concentrate on calculating all the joint probabilities using less processing time. For example, an ad hoc approach for determining probabilities that approximate the calculations of the JPDA technique is proposed in [2]. This approach is fast and easy to implement at the expense of lower overall performance. The issue of approximating the JPDA calculations can also be addressed using an optical processor, a depth-first search (DFS) mathematical method, a distributed processor method, and an algorithm using matrix manipulations [1,2,7,13–15].

3. Joint probabilistic data association technique

Because the proposed approach is based on the JPDA filter, a quick review of the JPDA technique is presented in this section. In the JPDA technique, joint probabilities are calculated over all targets and all hits. Each joint probability represents the probability of all the individual events combined. Thus JPDA includes the probability of multiple hits in one trajectory gate coming from multiple targets.

A detailed derivation of the JPDA equations can be found in many studies [12,16] and is briefly mentioned here. The equations of the JPDA technique have the same form as standard Kalman filter equations with the following two changes:

- (1) The state estimate of the i th target is updated according to

$$\hat{\mathbf{x}}_i(k+1|k+1) = \hat{\mathbf{x}}_i(k+1|k) + \mathbf{K}_i(k+1) \sum_{j=1}^{n_p} \beta_{ij} \tilde{\mathbf{z}}_{ij}(k+1), \quad i = 1, 2, \dots, n_t, \quad (12)$$

where $\tilde{\mathbf{z}}_{ij}(k+1)$ is the residual (innovation) due to hit j and target i , β_{i0} is the *a posteriori* probability that no hit originated from target i , and $\beta_{ij}, \forall j \neq 0$, is the *a posteriori* probability that a validated hit j originated from target i .

- (2) The update covariance matrix is evaluated as

$$\mathbf{P}_i(k+1|k+1) = \mathbf{P}_i^o(k+1|k+1) + d\mathbf{P}_i(k+1|k+1), \quad (13)$$

where $\mathbf{P}_i^o(k+1|k+1)$ and $d\mathbf{P}_i(k+1|k+1)$ are defined as

$$\mathbf{P}_i^0(k+1|k+1) = \beta_{i0}\mathbf{P}_i(k+1|k) + (1 - \beta_{i0})[\mathbf{I} - \mathbf{K}_i(k+1)\mathbf{H}_i(k+1)]\mathbf{P}_i(k+1|k), \quad (14)$$

$$d\mathbf{P}_i(k+1|k+1) = \mathbf{K}_i(k+1) \left[\sum_{j=1}^{n_p} \beta_{ij} \tilde{\mathbf{z}}_{ij}(k+1) \tilde{\mathbf{z}}'_{ij} - \tilde{\mathbf{z}}_i(k+1) \tilde{\mathbf{z}}'_i(k+1) \right] \mathbf{K}'_i(k+1). \quad (15)$$

Expressions for the *a posteriori* probabilities, in the case of the JPDA technique, are derived in [12–16] in the case of Gaussian distributed observations.

The idea of the proposed association approach is to use possibility weights instead of probability weights. This is considered in many studies [7,51,56]. The theory of possibility and the concept of possibility–probability consistency are also addressed in many studies [1,7,51,56,59,63]. The Fuzzy Subsethood theorem addressed these important issues [34]. The results of this work showed that the Fuzzy Subsethood theorem implies Bayes theorem or, equivalently, probability represents a special case of possibility. The comparison between probability and possibility shows that the results of the fuzzy set theorem are inferior to those of the probability theorem [51,56]. This conclusion is expected because probability theorem utilizes a priori information about the environment. In general, probability weights can be replaced by possibility weights. This depends on the available information and the type of application [1,7,51]. Possibility is considered to be an upper bound for probability [59,63]. The possibility–probability consistency principle is established by Zadeh [59]. The degree of non-contradiction, i.e., the consistency between a possibility distribution $\pi_X(x)$ and a probability distribution $f_X(x)$, is represented by a consistency measure [59]

$$C(\pi_X, f_X) = \pi_1 p_1 + \pi_2 p_2 + \cdots + \pi_n p_n = \sum_{x \in X} \pi_X f_X, \quad (16)$$

where $f_X(x)$, $\pi_X(x)$, and $C(\pi_X, f_X)$ are values in the range [0,1].

Our objective in this paper is to propose a multitarget tracking data association technique based on possibility weights instead of probability weights that not only performs well, but is also feasible for real time implementation of multitarget tracking systems.

4. Proposed joint possibilistic data association technique

Multitarget tracking is the most important application for most surveillance systems. The main objective of a tracking system is the estimation of target trajectories and their prediction into the near future. Estimation problem involves the attempt to find the value of a state vector (e.g. position, velocity, etc.) that best fits, in a defined mathematical sense, the observational data. We deal with dynamic estimation problems, in which the state vector changes as a function of time. Having performing hits-to-targets association, then estimation techniques can be used to improve the estimate of the state of the targets. The processes of association and estimation are interleaved and performed concurrently. Thus data association is necessary to allow estimation techniques to be utilized, and estimation techniques provide a mean of updating the state estimates used in the association process.

The key to successful target tracking lies in the effective extraction of useful information about the observed target's states from the received hits. When tracking multiple targets in a cluttered environment, a batch of hits is received at each sampling time (scan) without a priori information on origin of each hit. In this case, the targets compete for hits and there is always uncertainty associated with the received hits as well as their origin. Hits may be false or clutter, or they may be data from unknown targets whose trajectories are described in a statistical manner. The objective of any target tracking is to process the received targets hits from a field of view containing one or more potential targets of interest and to then partition the received hits into sets or tracks that are produced by the same target. Once trajectories (tracks) are formed and confirmed, the observed targets of interest can be estimated. Then quantities, such as velocity and predicted position, can be computed for each target track. Since most.

When multiple targets as well as clutter are present in the same neighborhood, the data association becomes much more complicated. For example, when tracking multiple targets using remote sensing devices, such as radars, sonars, infrareds, and electro-optical devices, the origin of hits is uncertain. In these cases, the remote sensing devices yield hits whose origin is uncertain, that is, not necessarily the targets of interest. This uncertainty occurs when a received hit from a target is weak and, to detect it, the detection threshold has to be lowered. In such cases, background hits and sensor noise may also be detected, yielding spurious hits (or clutter). This situation can also occur when multiple targets are present in the same neighborhood or in case of crossing targets. In a cluttered environment, the correct hit is not known for each target trajectory and the problem is how to select the true hit from all the received hits. Using the clutter in a tracking system leads to divergence of the estimation error and, thus, track loss.

Because the proposed approach depends on possibility weights, it is denoted by DAPW (data association based on possibility weights). We assume the case of a multiple-target tracking system with n_t target trajectories in the presence of clutter. Clutter refers to random received hits from nearby targets, false alarms, interference, weather, etc. The spatial density of the clutter hits is assumed to be λ . Assume that the validation gates of the targets under surveillance are defined in some way, and n_v hits are found inside these gates. The validation gate of a target is defined as the region in which the true hit will appear with high probability. The gate probability of the i th target is denoted by P_{Gi} .

Multidimensional gates are then set up for detecting the hits from the targets. This step avoids searching for the hits from the targets of interest in the entire hit space. However, hits in the gates, which not guaranteed to have originated from the targets associated with the gates, are valid association candidates, and such gates are called validation gates. The validation regions are set up to ensure that the targets hits fall in the validation regions with high probabilities, called gate probabilities, based on the statistical characterization of the predicted hits. Hits outside the validation regions can be ignored because they are too far from the predicted hits and thus unlikely to have originated from the targets of interest. All the hits in the validation regions may have originated from the targets of interest, even though at most one is the true one. Consequently, there are many possible association events. If more than one hit appears in a gate, then we face an association uncertainty, namely, we must determine which hit is target originated and thus to be used to update the target track, which consists of the state estimate and covariance matrix. When estimating the states of multiple targets using hits with uncertain origin, it is not known which hit originated from which target or from clutter.

In the nearest neighbor data association, the hit satisfactorily closest to the predicted one is used in the estimation process, i.e. the nearest neighbor association declares the hit that is closest to the predicted target location to be the true target detection. The nearest neighbor association approaches assume that the hit closest to the center of the validation gate represents the target. All other hits in the validation gate are then labeled as false alarms. Such simple association techniques are rarely effective in case of multitargets in clutter or in case of closing and crossing targets. The primary disadvantage is that clutter can pull the tracker away from the target position. Instead of using only one hit for each target, among all of the validated hits, and discarding the others, the proposed joint possibilistic data association approach use all of the validated hits with different weights. In JPDA approach, the weights are probability weights, while in the proposed approach, the weights are possibility weights. The possibility weights are determined using the noise covariance matrices and the current received hits such that the total sum of the distances between all hits and targets is minimized. The objective here is to obtain in real-time the possibility that each validated hit is attributable to the target of interest such that the mean square error estimate of the target state and the associated uncertainty is minimized. The proposed data association based on possibility weights consists of two main steps; determination of the possibility weights and updating of the state estimate and the covariance matrix. These two main steps are illustrated in this section.

The set of validated hits, n_v , includes the true target hits (when targets are detected) and false target hits originating from clutter. The detection probability of the i th target is denoted by pd_i , $i = 1, 2, \dots, n_t$. It represents the probability that the correct hit is detected at all.

The validation gates of the targets indicate the possible joint events. We define a validation matrix as:

$$\Omega = [\omega_{ij}], \quad i = 1, 2, \dots, n_t, \quad j = 0, 1, 2, \dots, n_v, \quad (17)$$

where ω_{ij} are binary elements (0 or 1). If hit S_p lies in the validation gate of target i , the corresponding binary element ω_{ij} is 1, otherwise, it will be 0. Thus,

$$\omega_{ij} = \begin{cases} 1, & \text{if hit } j \in \text{target } i \\ 0, & \text{otherwise.} \end{cases} \quad (18)$$

The index $j = 0$ in (17) represents the case in which none of the hits is valid. A target may be assigned hit 1, hit 2, etc., or hit n_v , or no assignment at all. Because all targets have the possibility of no valid hits, the first column of the validation matrix Ω has all units, i.e. $\omega_{ij} = 1$, $i = 1, 2, \dots, n_t$, $j = 0$.

We explain how to derive the validation matrix Ω with an example. Suppose that there are four targets ($n_t = 4$), and nine hits ($n_v = 9$) fall within the validation gates of the four targets. Fig. 1 illustrates the association problem that occurs in case of only four targets and nine hits. It shows that hit 1 falls within the validation gate of target 1 ($\omega_{11} = 1$), outside the validation gate of target 2 ($\omega_{21} = 0$), outside the validation gate of target 3 ($\omega_{31} = 0$), and outside the validation gate of target 4 ($\omega_{41} = 0$). Similarly, we can obtain all values of the binary elements of the validation matrix $\Omega = [\omega_{ij}]$, $i = 1, 2, 3, 4$, $j = 0, 1, 2, \dots, 9$. The binary values $\omega_{10} = 1$, $\omega_{20} = 1$, $\omega_{30} = 1$, and $\omega_{40} = 1$ represent the cases in which none of the nine hits is valid for target 1, target 2, target 3, and target 4, respectively.

The validation matrix of the example depicted in Fig. 1 is

$$\Omega(i, j) = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}, \quad i = 1, 2, 3, 4, \quad j = 0, 1, 2, \dots, 9. \quad (19)$$

To quantify the association between hits-to-track pairs, an association metric must be defined. In the proposed association approach, the association metric (distance measure) Δ_{ij} between hit j and target i is defined in terms of the norm of the residual d_{ij}^2 , where the squared distance d_{ij}^2 is the weighted inner product of the innovation vector between hit j and target i , i.e.,

$$d_{ij}^2 = \tilde{\mathbf{z}}_j' \mathbf{S}_i^{-1} \tilde{\mathbf{z}}_j = [\mathbf{z}_j - \hat{\mathbf{z}}_i]' \mathbf{S}_i^{-1} [\mathbf{z}_j - \hat{\mathbf{z}}_i], \quad (20)$$

where the innovation $\tilde{\mathbf{z}}_{ij} = [\mathbf{z}_j - \hat{\mathbf{z}}_i]$ is the vector difference between the measured and the predicted quantities, and $\mathbf{S}_i$ is the covariance of the innovation given by (11). Weighting factors can also be applied to scale effectively the components of the candidate hits. When the inverse covariance matrix is used as a weight, as in Eq. (20), the resulting association metric is called nearest neighbor standard filter (NNSF). Based on the validation matrix Ω , the proposed association metric $\{\Delta_{ij}\}$ are weighted by other factors according to the following rules:

- (1) Since a hit falling within a gate is more likely from the trajectory in question than from an extraneous source and the probability that a hit is a valid return is directly proportional to the detection probability, the factors $(pd_i P_{Gi})$ and $(1 - pd_i P_{Gi})$ should be included in the definition of Δ_{ij} to represent the probabilities of the target being detected ($\omega_{ij} = 1$) and not detected ($\omega_{ij} = 0$) taking the gate probability of target i (P_{Gi}) into consideration. In the special case where $pd_i = 1$ and $P_{Gi} = 1$, there are no expected extraneous returns and the validation gate corresponds to the entire surveillance region (the gate size should be infinity).
- (2) If $\omega_{ij} = 1, j \neq 0$, then hit j is detected by trajectory i and Δ_{ij} is directly proportional to the weighted inner product of the innovation vector between hit j and target i . This represents the case when hit j is inside the gate of trajectory i . In this case, the factor λ^{n_i-1} should be included in the definition of Δ_{ij} to represent that one of the hits in the validation gate of target i is valid and the remaining $(n_i - 1)$ hits will be due to clutter, where n_i is the number of validated hits within the gate of target i .
- (3) If $j = 0$, then the factor λ^{n_i} should be included in the definition of Δ_{ij} to represent the case in which none of the hits in the validation gate of target i is valid. In this case, all the n_i hits will be due to clutter.

Based on (20) and the previous rules, the association metric Δ_{ij} between hit j and target i is defined as:

$$\Delta_{ij} = \begin{cases} pd_i P_{Gi} d_{ij}^2 \lambda^{n_i-1}, & \forall j \neq 0, \omega_{ij} = 1, i = 1, 2, 3, \dots, n_t, \\ (1 - pd_i P_{Gi}) \lambda^{n_i}, & \forall \omega_{ij}, j = 0, i = 1, 2, 3, \dots, n_t. \end{cases} \quad (21)$$

Note that if $\omega_{ij} = 0, j \neq 0$, then hit j is not detected by track i and this means that Δ_{ij} is very large (theoretically ∞). This represents the case when measurement j is very far from track i (i.e. outside its gate).

Define Φ as a partition matrix of elements $\phi_{ij}, i = 1, 2, \dots, n_t, j = 0, 1, 2, \dots, n_v$, which represents the possibility that measurement j is a valid return for target i , such that:

$$\phi_{ij} = \Delta_{ij} = (1 - pd_i P_{Gi}) \lambda^{n_i}, i = 1, 2, 3, \dots, n_t, j = 0, \quad (22)$$

and

$$\begin{aligned} \phi_{ij} &\in [0, 1], 1 \leq i \leq n_t, 1 \leq j \leq n_v, \\ \sum_{i=1}^{n_t} \phi_{ij} &= 1 \quad \forall j \neq 0, \\ 0 &\leq \sum_{j=1}^{n_v} \phi_{ij} \leq n_t \quad i = 1, 2, \dots, n_t. \end{aligned} \quad (23)$$

Define S_q as the sum of the squared errors weighted by the q th power of the corresponding possibility, i.e.

$$S_q = \sum_{j=1}^{n_v} \sum_{i=1}^{n_t} (\phi_{ij})^q \Delta_{ij}, i = 1, 2, \dots, n_t, j = 1, 2, \dots, n_v. \quad (24)$$

For any given $\Delta_{ij} \forall i, j \neq 0$ and q , it is required to obtain the correlations (association) between all hits and all targets $\{\phi_{ij}, \forall i, j \neq 0\}$, such that the total sum of the square errors S_q is minimized. This is a squared error criterion and can be solved using Lagrange multipliers. The solutions for the correlations between all hits and all targets, $\phi_{ij}, \forall i, j \neq 0$, are given by [4,34,63]:

$$\phi_{ij} = \frac{(1/\Delta_{ij})^{1/(q-1)}}{\sum_{l=1}^{n_v} (1/\Delta_{il})^{1/(q-1)}}, i = 1, 2, \dots, n_t, j = 1, \dots, n_v. \quad (25)$$

Note that the correlation weights in Eq. (25) depend on the integer power, q . A larger integer power means smaller correlation weights [21,34]. The value of q reduces the influence of small correlations compared to that of large correlations. If $q = 2$, the values of the correlation weights in Eq. (25) are determined by a weighted Euclidean norm (see Eqs. (20) and (21)). In this case, the expressions of the correlation weights in Eq. (25) have similar forms to the standard association approach, which depends on the chi-squared metric, i.e., they depend on the weighted inner products of the innovations [12–16]. Thus, we choose $q = 2$ in all simulated examples.

From (25), it is clear that if $\Delta_{ij} = \infty$, the corresponding correlation ϕ_{ij} will be 0. The correlation values $\phi_{ij}, \forall i, j$ are normalized to be $p_{ij}, \forall i, j$, such that, for a given trajectory i , the contribution of all hits is equal to unity, i.e.,

$$\sum_{j=0}^{n_v} p_{ij} = 1, i = 1, 2, \dots, n_t. \quad (26)$$

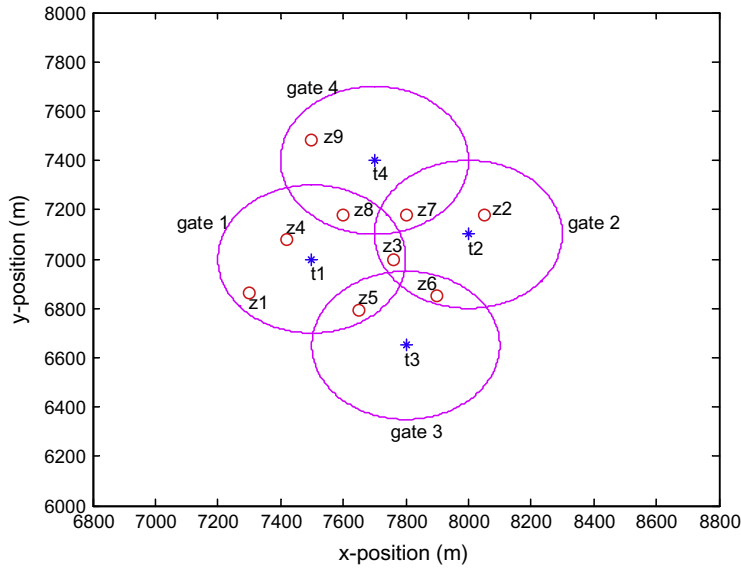


Fig. 1. An example of four targets and nine measurements.

Each value p_{ij} in Eq. (26) represents the possibility that hit j is a valid hit for target i . The possibility values $p_{ij}, \forall i, j$ are used to determine the targets' state estimates, the updated covariance matrices, and the state updated equations. We use the same equations of the JPDA technique except that the probability weights between all targets and hits $\{\beta_{ij}, \forall i, j\}$ are replaced by the possibility weights between all targets and hits $\{p_{ij}, \forall i, j\}$.

The i th target state estimate is updated based on the new hits according to

$$\hat{\mathbf{x}}_i(k+1|k+1) = \hat{\mathbf{x}}_i(k+1|k) + \mathbf{K}_i(k+1) \sum_{j=1}^{n_v} p_{ij} \tilde{\mathbf{z}}_{ij}(k+1), \quad i = 1, 2, \dots, n_t, \quad (27)$$

where $\tilde{\mathbf{z}}_{ij}(k+1)$ is the residual error due to hit j and target i at time instant $k+1$ and is given by

$$\tilde{\mathbf{z}}_{ij}(k+1) = \mathbf{z}_j(k+1) - \hat{\mathbf{z}}_i(k+1), \quad (28)$$

and $\mathbf{K}_i(k+1)$ is the Kalman filter gain for target i .

The state update equation for target i can be written as

$$\hat{\mathbf{x}}_i(k+1|k+1) = \hat{\mathbf{x}}_i(k+1|k) + \mathbf{K}_i(k+1) \tilde{\mathbf{z}}_i(k+1), \quad (29)$$

where

$$\tilde{\mathbf{z}}_i(k+1) = \sum_{j=1}^{n_v} p_{ij} \tilde{\mathbf{z}}_{ij}(k+1) \quad (30)$$

is the sum of weighted innovations. The update covariance matrix for target i is evaluated as

$$\mathbf{P}_i(k+1|k+1) = \mathbf{P}_i^o(k+1|k+1) + d\mathbf{P}_i(k+1|k+1), \quad (31)$$

where $\mathbf{P}_i^o(k+1|k+1)$ is the Kalman filter covariance that would be calculated if a single hit were present. The term $d\mathbf{P}_i(k+1|k+1)$ is added to reflect the effect of uncertain correlation for target i . The two equations determining $\mathbf{P}_i^o(k+1|k+1)$ and $d\mathbf{P}_i(k+1|k+1)$ are

$$\mathbf{P}_i^o(k+1|k+1) = p_{i0} \mathbf{P}_i(k+1|k) + (1 - p_{i0})[\mathbf{I} - \mathbf{K}_i(k+1)\mathbf{H}_i(k+1)]\mathbf{P}_i(k+1|k), \quad (32)$$

$$d\mathbf{P}_i(k+1|k+1) = \mathbf{K}_i(k+1) \left[\sum_{j=1}^{n_v} p_{ij} \tilde{\mathbf{z}}_{ij}(k+1) \tilde{\mathbf{z}}_{ij}'(k+1) - \tilde{\mathbf{z}}_i(k+1) \tilde{\mathbf{z}}_i'(k+1) \right] \mathbf{K}_i'(k+1). \quad (33)$$

The term $d\mathbf{P}_i(k+1|k+1)$ in Eq. (33) increases the covariance matrix according to the possibility weights as well as the spread of all the target hits.

Note that the choice of d_{ij}^2 in (21) determines the shape of the target spreading. If we choose the usual Euclidean form, i.e., $d_{ij}^2 = \tilde{\mathbf{z}}_{ij}' \tilde{\mathbf{z}}_{ij} = [\mathbf{z}_j - \hat{\mathbf{z}}_i]' [\mathbf{z}_j - \hat{\mathbf{z}}_i]$, the shape of target spreading will be an equally sized hypersphere. We can also choose the

diagonal form $d_{ij}^2 = \tilde{\mathbf{z}}_{ij} [\text{diag}(\sigma_k^2)]^{-1} \tilde{\mathbf{z}}_{ij}$, where σ_k^2 denotes the variance of the feature k , i.e., $d_{ij}^2 = \tilde{\mathbf{z}}_{ij} \mathbf{R}^{-1} \tilde{\mathbf{z}}_{ij}$, or we can choose the Mahalanobis form for which $d_{ij}^2 = \tilde{\mathbf{z}}_{ij} \mathbf{P}^{-1} \tilde{\mathbf{z}}_{ij}$ [12,14,15].

In summary, the proposed approach consists of the following steps:

- (1) Given n_t target tracks with clutter spatial density of λ . Assume validation gates of the targets with corresponding $\{P_G\}$ are defined in some way (see [12–16]), and n_p hits are found inside them. The target detection probabilities are assumed to be $\{pd\}$.
- (2) Construct the validation matrix $\Omega = [\omega_{ij}]$, $i = 1, 2, \dots, n_t$, $j = 0, 1, 2, \dots, n_p$, using Eq. (18).
- (3) Based on the validation matrix Ω , construct the partition matrix Φ using Eqs. (22) and (25).
- (4) Normalize the possibility scores $\{\varphi_{ij}, \forall i, j\}$ using Eq. (26).
- (5) Update the state estimate and covariance matrix according to the standard Kalman filter Eqs. (7) and (8) with the following two changes:
 - (i) The state estimate of the i th target is updated according to

$$\hat{\mathbf{x}}_i(k+1|k+1) = \hat{\mathbf{x}}_i(k+1|k) + \mathbf{K}_i(k+1) \sum_{j=1}^{n_p} p_{ij} \tilde{\mathbf{z}}_{ij}(k+1), \quad i = 1, 2, \dots, n_t, \quad (34)$$

Table 1

Expressions for the total number of inner products in the case of the NNSF technique, the JPDA technique, and the proposed DAPS technique.

Data association approach	Total number of inner products (n_p)
NNSF	$n_t n_p$
JPDA	$\left(1 + \sum_{n=0}^{n_t-1} \frac{n_t! n_p!}{n! (n_t-n)! (n_p-n)!}\right) (n_t - 1)$
Proposed DAPS	$n_t (n_p + 1)$

Table 2

Comparison of the total number of inner products in the case of the NNSF technique, the JPDA technique, and the proposed DAPS technique.

n_t	n_p	n_p In case of NNSF	n_p In case of JPDA	n_p In case of the Proposed DAPS
2	5	10	31	12
3	7	21	716	24
4	9	36	16,527	40
5	10	50	254,364	55
6	14	84	20,052,275	90
7	16	112	664,060,998	119
8	19	152	3.9978×10^{10}	160

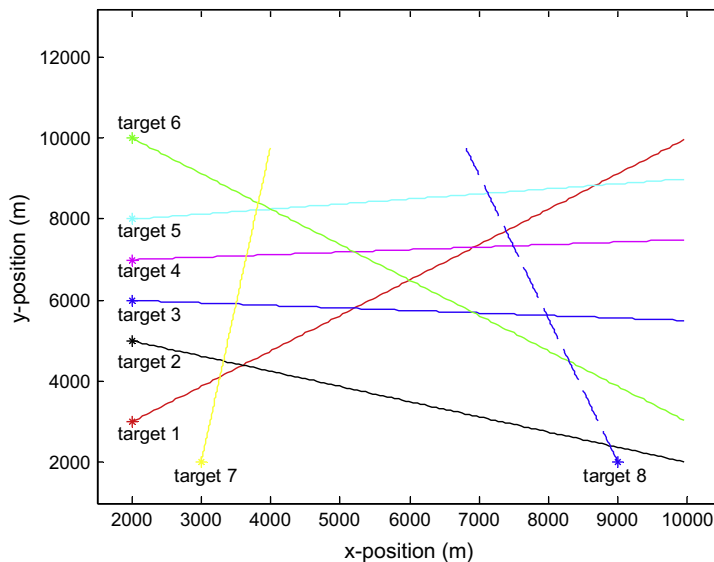


Fig. 2. True trajectories of the eight targets.

where each possibility score p_{ij} , $\forall j \neq 0$ represents the possibility that a validated hit j originated from target i and is determined according to Eqs. (25) and (26).

(ii) The update covariance matrix is evaluated as

$$\mathbf{P}_i(k+1|k+1) = \mathbf{P}_i^o(k+1|k+1) + d\mathbf{P}_i(k+1|k+1), \quad (35)$$

where

$\mathbf{P}_i^o(k+1|k+1)$ and $d\mathbf{P}_i(k+1|k+1)$ are defined as

$$\mathbf{P}_i^o(k+1|k+1) = p_{i0}\mathbf{P}_i(k+1|k) + (1-p_{i0})[\mathbf{I} - \mathbf{K}_i(k+1)\mathbf{H}_i(k+1)]\mathbf{P}_i(k+1|k), \quad (36)$$

$$d\mathbf{P}_i(k+1|k+1) = \mathbf{K}_i(k+1) \left[\sum_{j=1}^{n_v} p_{ij} \tilde{\mathbf{z}}_{ij}(k+1) \tilde{\mathbf{z}}_{ij}'(k+1) - \tilde{\mathbf{z}}_i(k+1) \tilde{\mathbf{z}}_i'(k+1) \right] \mathbf{K}_i'(k+1). \quad (37)$$

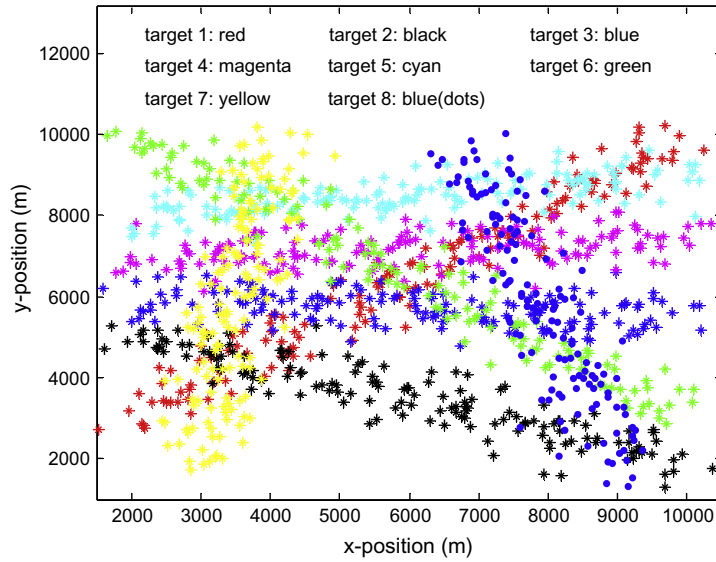


Fig. 3. Received noisy measurements of the eight targets.

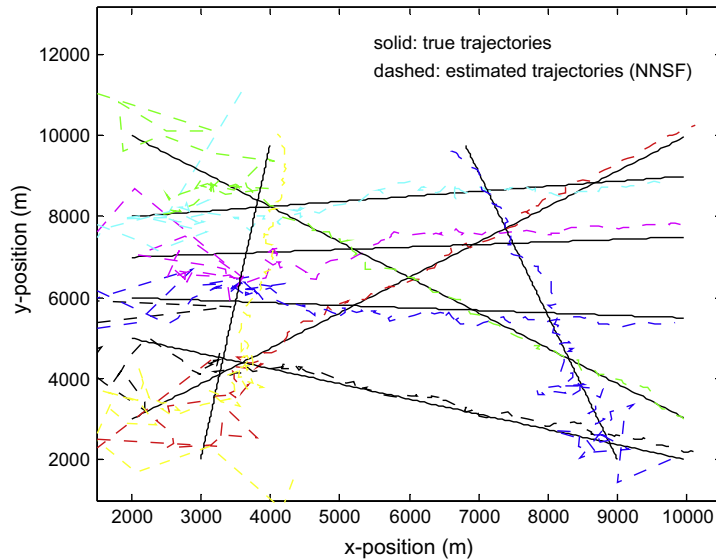


Fig. 4. True and estimated trajectories in the case of the NNSF technique.

The possibility score p_{i0} represents the possibility that no hit originated from target i , and is evaluated according to Eqs. (22) and (26).

5. Comparison of the computational requirements

In the JPDA technique, the probability weights are determined over all hits and all targets by considering all feasible hypotheses [12,16]. The total number of feasible hypotheses varies according to the scenario represented by the validation matrix Ω . A special case exists when all hits fall inside the intersection of the validation gates of all targets. This is the worst possible case in which each hit may be associated with either clutter or any of the n_t targets. In this worst case, all of the binary elements of the validation matrix have the unity value, i.e., $\omega_{ij} = 1$, $i = 1, 2, \dots, n_t$, $j = 0, 1, \dots, n_v$. An expression for the total number of feasible hypothesis is presented in [1,25] for this special case. The results of these papers showed that the total number of feasible hypotheses increases exponentially with the total number of targets n_t and the total number of validated hits n_v . This method is impractical for real time processing in the case of a large number of targets and hits.

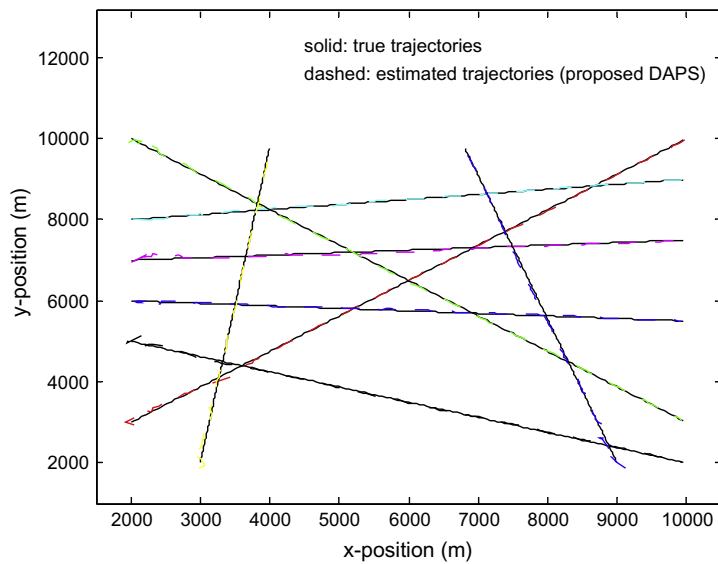


Fig. 5. True and estimated trajectories in the case of the proposed DAPS technique.

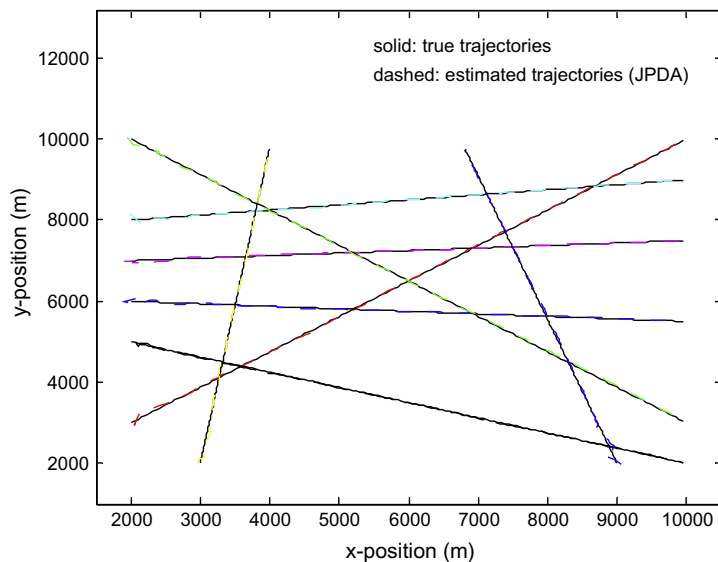


Fig. 6. True and estimated trajectories in the case of the JPDA technique.

In multitarget tracking systems, the required computations are determined by the total number of inner products. In the worst case of the JPDA technique, the total number of inner products (n_p) is given by [1,14,15,25]

$$n_p = \left(1 + \sum_{n=0}^{n_t-1} \frac{n_t! n_v!}{n! (n_t - n)! (n_v - n_t + n)!} \right) (n_t - 1). \quad (38)$$

The previous equation holds when the number of hits is greater than the number of targets (i.e., the practical case). When the number of targets is greater than the number of hits, the values of n_v and n_t are interchanged in Eq. (38).

In the case of the NNSF technique, the correlation between all targets and all hits requires computing $n_t n_v$ inner products [12–15]. In the proposed DAPS approach, the total number of inner products for determining the possibility weights over all targets and all hit, depends on a possibility matrix of dimension $n_t \times (n_v + 1)$. Therefore, the total number of inner products in the case of DAPS is $n_t(n_v + 1)$. Table 1 summarizes the expressions for the total number of inner products in the case of JPDA, NNSF, and the proposed DAPS method. Table 2 compares some numerical values of the total number of inner products (n_p) for some values of n_t and n_v in the case of NNSF, JPDA, and the proposed DAPS method. Tables 1 and 2 show that the

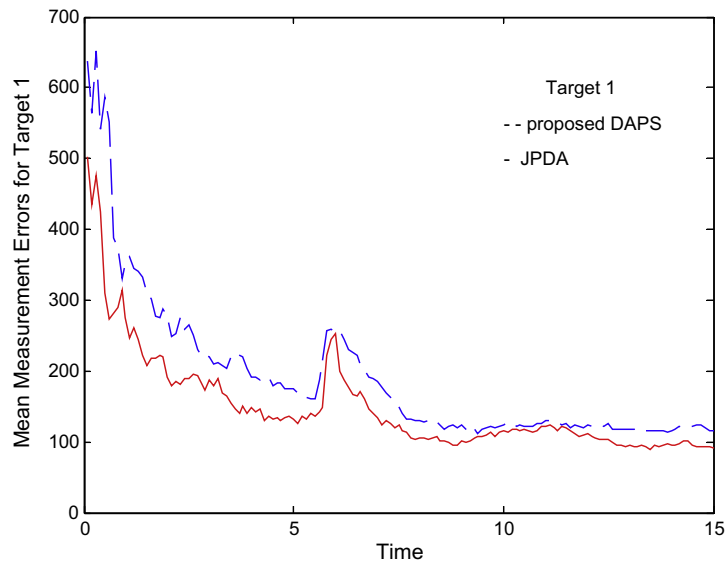


Fig. 7. Comparison of the JPDA technique and the proposed DAPS technique for target 1.

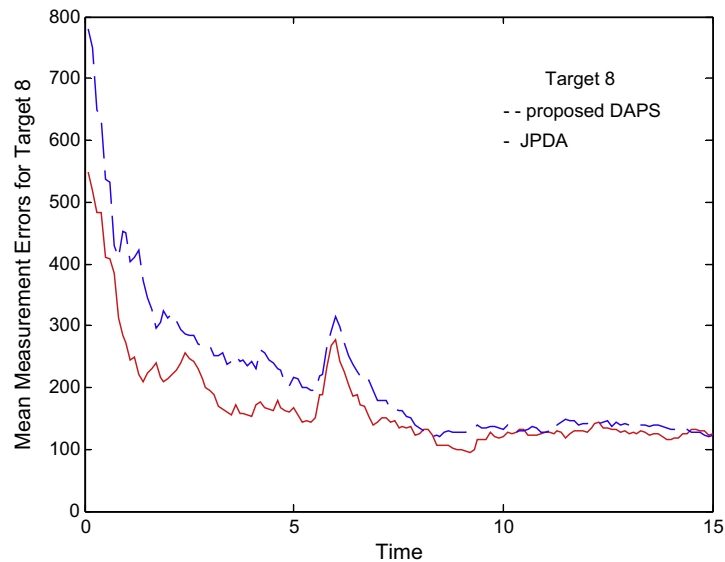


Fig. 8. Comparison of the JPDA technique and the proposed DAPS technique for target 8.

computational requirements of the JPDA technique are quite high and infeasible in the case of a large number of hits and targets. Tables 1 and 2 also show that the proposed DAPS method has fewer computations compared to the JPDA technique and has comparable computations compared to the NNSF technique.

6. Performance comparison

6.1. A case of targets moving in straight lines

First, we consider a case of targets moving in straight lines. We consider a two-dimensional tracking system with eight crossing targets in a cluttered environment. The eight targets are assumed to move in straight lines without acceleration, i.e., with constant velocities. The received measurements are created by simulating the true targets' trajectories in two

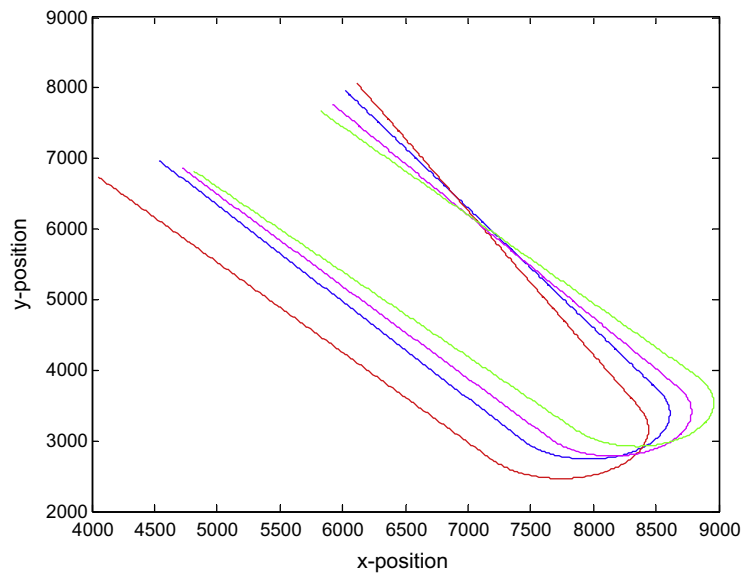


Fig. 9. True trajectories of the four maneuvering targets.

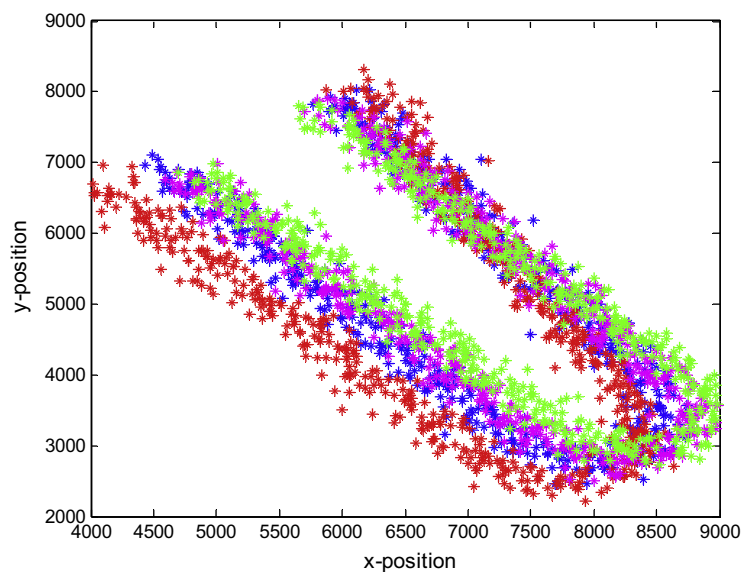


Fig. 10. Received noisy measurements of the four maneuvering targets.

dimensions (x and y), adding noise to the true targets' trajectories, and then adding clutter hits. The targets' motion models are defined by Eq. (1) with the state transition matrix $\mathbf{F}$ given by [1,7,12–16]

$$\mathbf{F} = \begin{pmatrix} 1 & \delta & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \delta \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (39)$$

where δ is the sampling interval.

The state vector, $\mathbf{x}(k)$, contains the x and y positions and velocities of each target, i.e.,

$$\mathbf{x}(k) = \begin{pmatrix} x(k) \\ v_x(k) \\ y(k) \\ v_y(k) \end{pmatrix}. \quad (40)$$

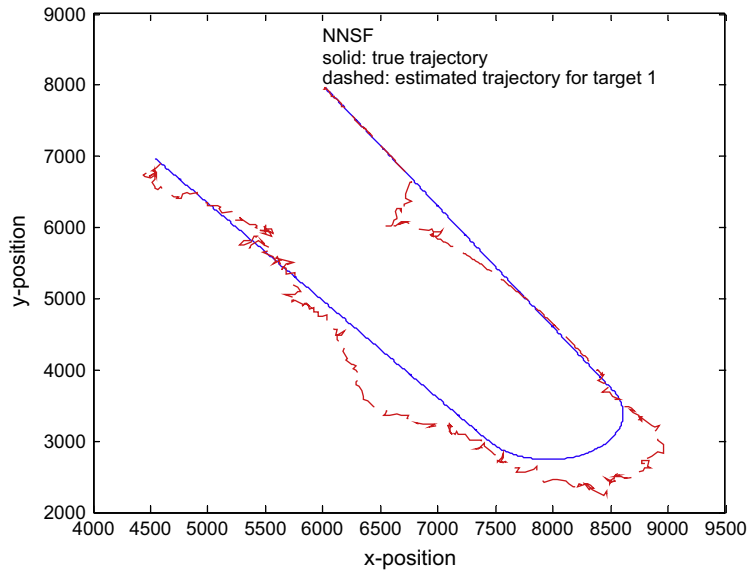


Fig. 11. True and estimated trajectories for target 1 with the NNSF technique.

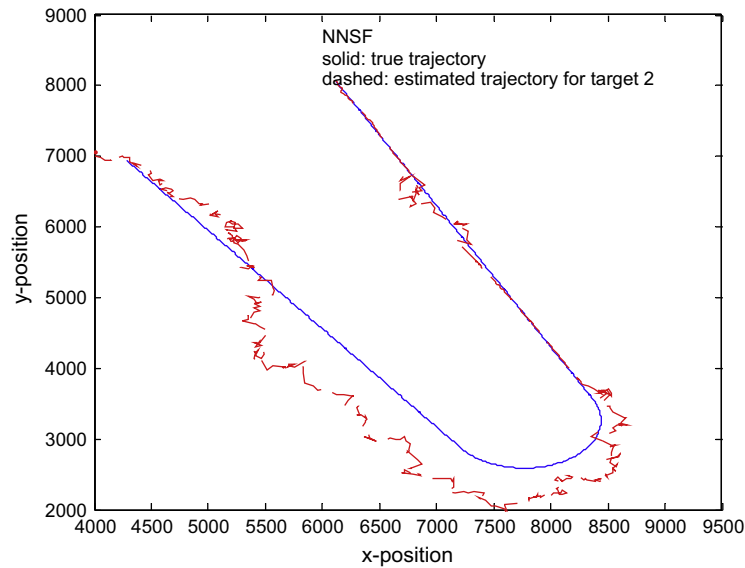


Fig. 12. True and estimated trajectories for target 2 with the NNSF technique.

The measurement matrix $\mathbf{H}$ is defined such that the received measurements are the x and y target positions, i.e.,

$$\mathbf{H} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (41)$$

The received measurements are affected by zero mean Gaussian noise with a standard deviation σ . The covariance matrix of the noise sequence $\mathbf{w}(k)$ is given by

$$\mathbf{R} = \text{Cov}(\mathbf{w}(k)) = \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{pmatrix}. \quad (42)$$

The noise standard deviations of the eight targets are assumed to be $\sigma_{xi} = \sigma_{yi} = 35$ m, $i = 1, 2, 3, 4$, and $\sigma_{xi} = \sigma_{yi} = 40$ m, $i = 5, 6, 7, 8$. The measurement error in both x and y positions is defined as

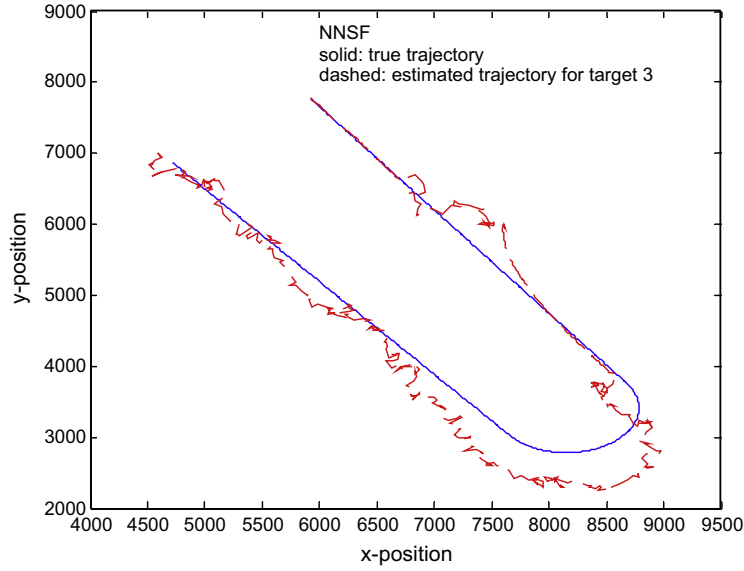


Fig. 13. True and estimated trajectories for target 3 with the NNSF technique.

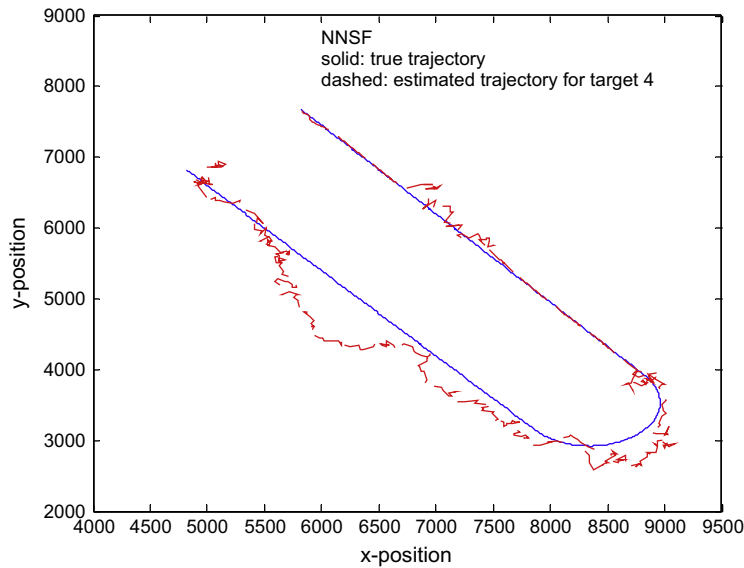


Fig. 14. True and estimated trajectories for target 4 with the NNSF technique.

$$e = \sqrt{e_x^2 + e_y^2} = \sqrt{(\mathbf{x}_{true} - \hat{\mathbf{x}})^2 + (\mathbf{y}_{true} - \hat{\mathbf{y}})^2}, \quad (43)$$

where $\mathbf{x}_{true}$ and $\mathbf{y}_{true}$ are the true target trajectories (unknown) while $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ are the estimated target trajectories using Kalman filter equations. The sampling interval is taken as 0.1 s and the targets' detection probabilities are taken as 1.0.

The clutter hits are assumed to be Poisson spatially distributed measurements with spatial density $\lambda = 0.0001$. The clutter hits are introduced in the tracking system at $t = 0$ s. We assume that the validation region for all targets is $P_G = 0.99$ as a standard gate to define the clutter density with a corresponding chi-square threshold of 9.2 (the two-dimensional case) [1,5,7]. A Monte Carlo simulation of 150 samples is conducted and results are obtained from 50 runs.

The true targets' trajectories (no process noise) are shown in Fig. 2 (unknown trajectories). The initial positions of the eight targets are marked by a '*' in Fig. 2. We add noise to the true targets' trajectories and then add clutter hits with clutter density λ . Fig. 3 shows the received noisy measurements for the eight targets. Fig. 4 shows the true trajectories as well as the estimated trajectories using the NNSF technique. In this case, the closest hit to the center of each target validation gate is

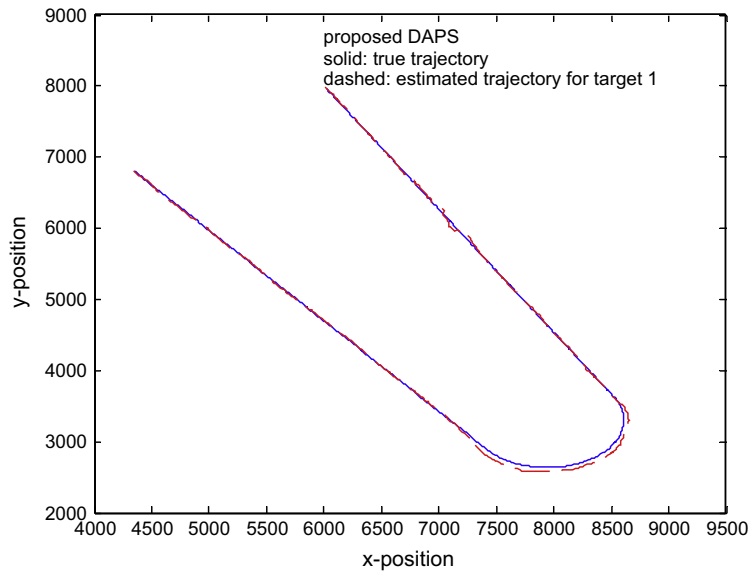


Fig. 15. True and estimated trajectories for target 1 with the proposed DAPS technique.

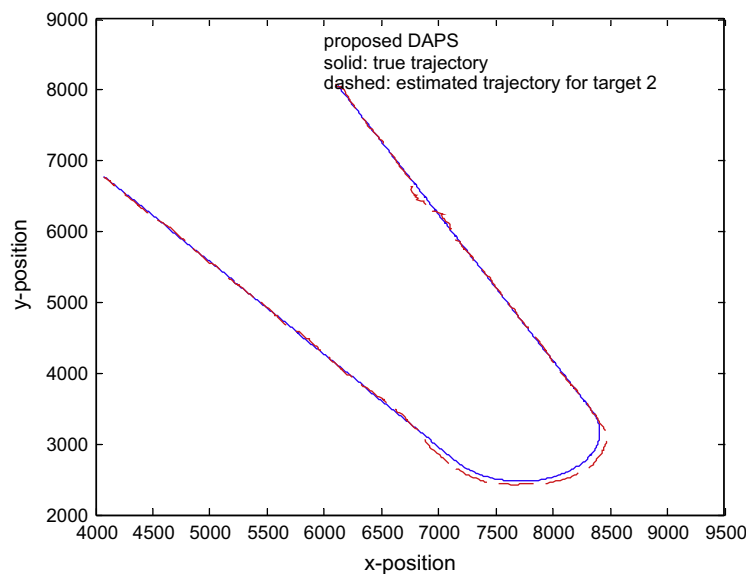


Fig. 16. True and estimated trajectories for target 2 with the proposed DAPS technique.

chosen as the correct hit. Fig. 4 shows that the estimated trajectories are far from the true trajectories. Thus, the NNSF technique is not accurate in this case. In the NNSF, the hit satisfactorily closest to the predicted one is used in the estimation process. Thus it assumes that the hit closest to the center of the validation gate represents the target. All other hits in the validation gate are then labeled as false alarms. As shown in Fig. 4, the clutter pulls the tracker away from the targets positions. This result is expected since the NNSF generally provides poor results in a crossing multitarget in a cluttered environment. Fig. 5 shows the estimated trajectories using the proposed DAPS technique. It is clear from Fig. 5 that the proposed DAPS approach improves the trajectories' estimates and outperforms the NNSF technique. To provide the ultimate performance, the estimated trajectories using the JPDA technique are shown in Fig. 6. Figs. 7 and 8 compare the JPDA approach and the proposed DAPS method in terms of the average measurement error, given by Eq. (43), for target 1 and 8, respectively. The average measurement error is evaluated over 50 runs. Figs. 7 and 8 show that the performance of the proposed DAPS approach is inferior and close to the performance of the JPDA technique. However, the proposed DAPS approach greatly reduces the computational complexity, as explained in Section 5. The degree of consistency between the possibility weights

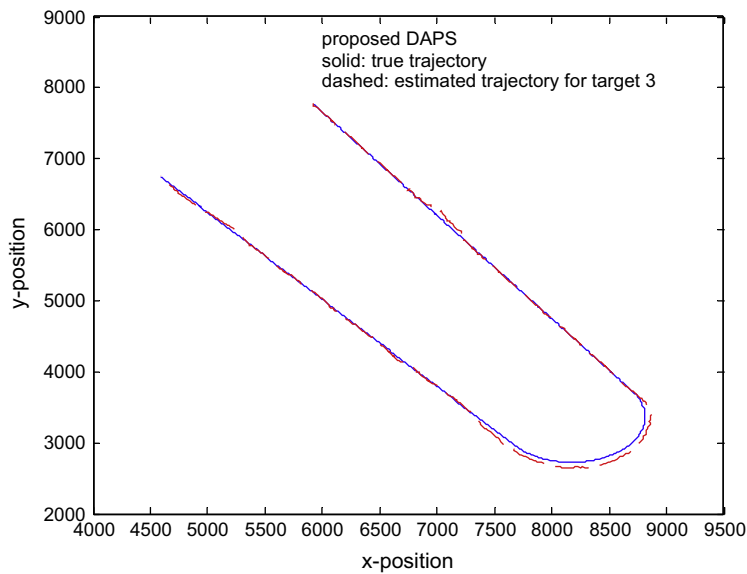


Fig. 17. True and estimated trajectories for target 3 with the proposed DAPS technique.

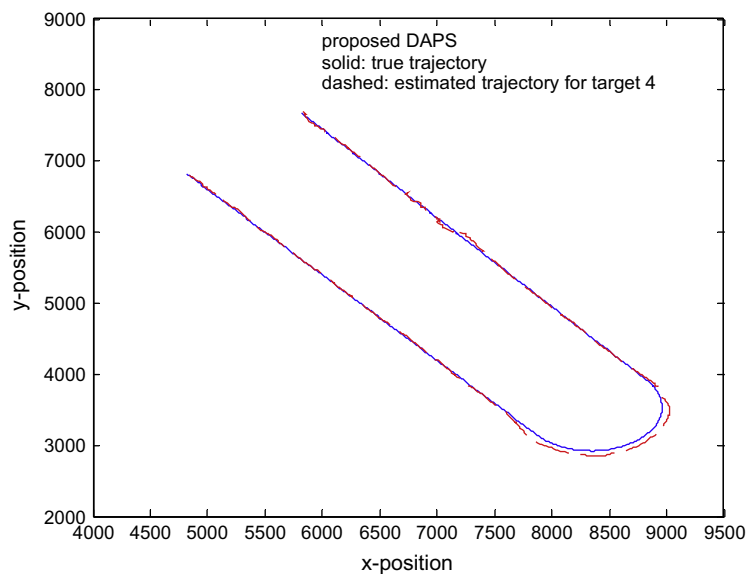


Fig. 18. True and estimated trajectories for target 4 with the proposed DAPS technique.

and the probability weights is calculated using Eq. (16) as an average of values over 150 samples and 50 runs and is found to be 0.864.

6.2. A case of maneuvering targets

Second, we consider a case of maneuvering targets. We consider a two-dimensional tracking system with four maneuvering targets in a cluttered environment. As before, the received measurements are created by simulating the true targets' trajectories in two dimensions (x and y positions), adding noise to the true targets trajectories, and then adding clutter hits. We assume that the dominant acceleration of the four maneuvering targets is the coordinated turn. Turning motion is usually the dominant target model for tracking systems. This is because the coordinated turn generates higher accelerations; up to ~ 9 g for an aircraft turn versus ~ 1 g for thrust. In addition, the targets usually maintain a high speed and turn rather than slowing down to avoid any danger. The targets' motion models have the same forms as Eqs. (1)–(4) with a covariance $\mathbf{Q}$ given by [1,13,15]:

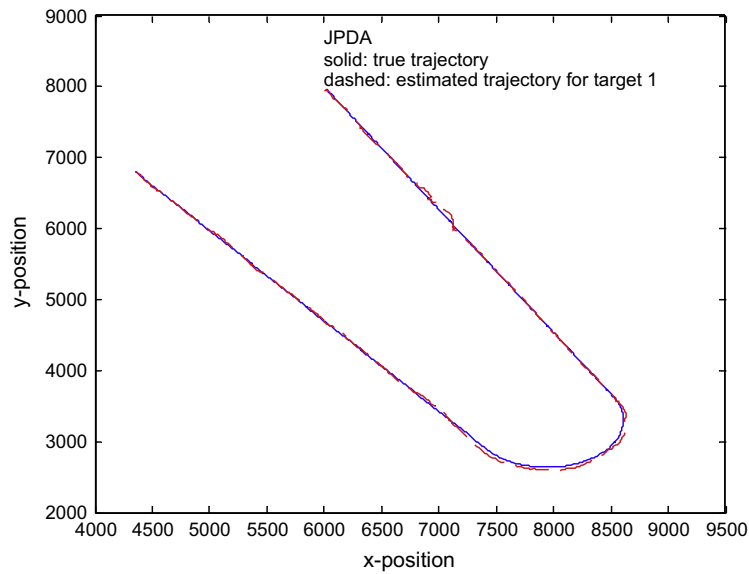


Fig. 19. True and estimated trajectories for target 1 with the JPDA technique.

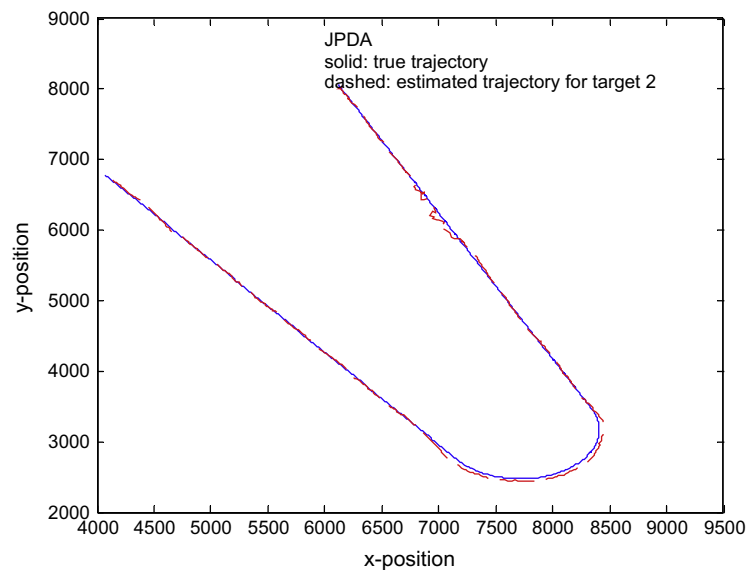


Fig. 20. True and estimated trajectories for target 2 with the JPDA technique.

$$\mathbf{Q} = a^2 \delta \begin{pmatrix} \delta^3/3 & \delta^2/2 & 0 & 0 \\ \delta^2/2 & \delta & 0 & 0 \\ 0 & 0 & \delta^3/3 & \delta^2/2 \\ 0 & 0 & \delta^2/2 & \delta \end{pmatrix}, \quad (44)$$

where a is the acceleration.

The initial states estimates and the corresponding initial covariance matrices are determined using measurements from two scans as explained in [1,7]. Each target motion model is initially assumed to be a straight line without acceleration. The hits are taken every 0.1 s. After generating 250 samples, i.e., after 25 s, each target executes a right turn with 10 g acceleration, where $g = 9.8 \text{ m}^2/\text{sec}$. Each target holds the turn for 100 samples and then returns again to straight line motion for an additional 250 samples. This simulation produces a total of 600 samples (60 s).

The noise standard deviations of the four maneuvering targets are assumed to be $\sigma_{x1} = \sigma_{y1} = 30 \text{ m}$, $\sigma_{x2} = \sigma_{y2} = 35 \text{ m}$, $\sigma_{x3} = \sigma_{y3} = 40 \text{ m}$, and $\sigma_{x4} = \sigma_{y4} = 45 \text{ m}$. Maneuvering target detection is performed using a method described in [13]. The values of

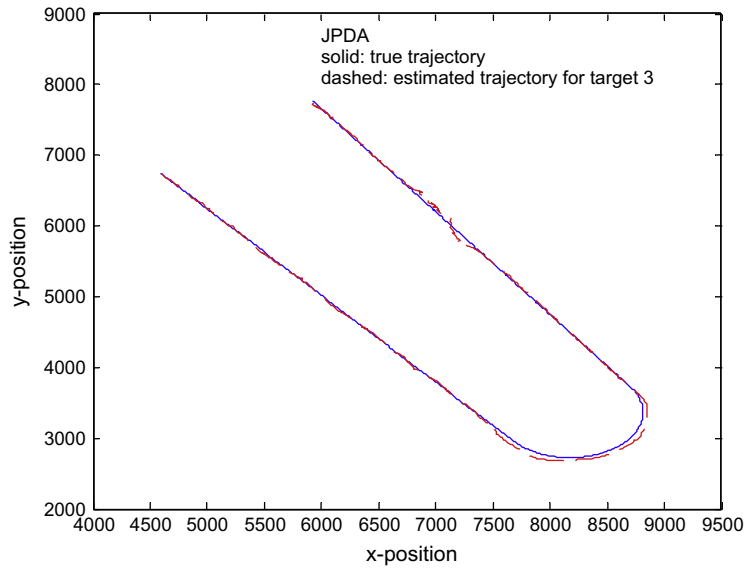


Fig. 21. True and estimated trajectories for target 3 with the JPDA technique.

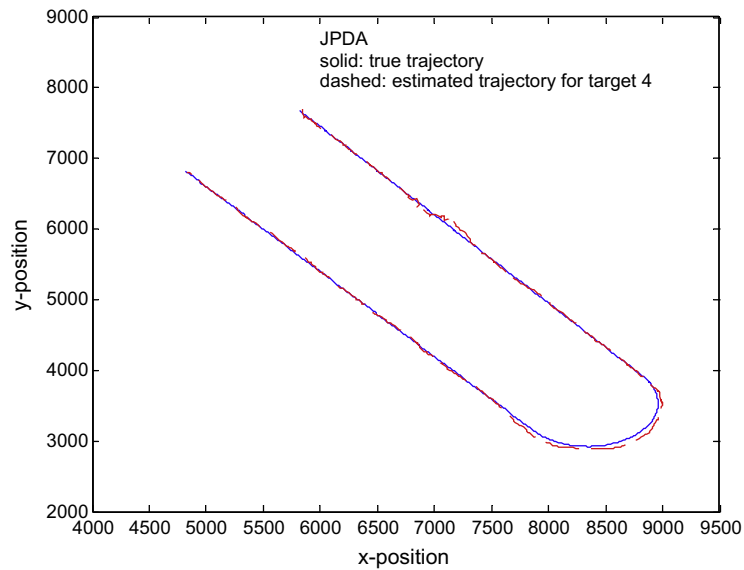


Fig. 22. True and estimated trajectories for target 4 with the JPDA technique.

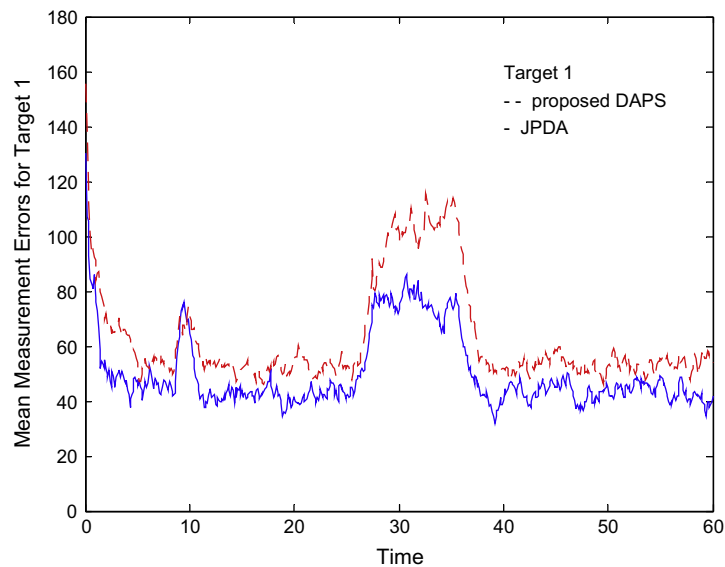


Fig. 23. Comparison of the JPDA technique and the proposed DAPS technique for target 1.

the gate thresholds, γ_i are taken from the χ^2 distribution table with two degrees of freedom, as presented in [1,7,13]. The tracking performance is evaluated based on 50 runs assuming that $\lambda = 0.0001$. The results are shown in Figs. 9–23. The true targets trajectories are shown in Fig. 9. The received noisy measurements for the four targets are shown in Fig. 10. Figs. 11–14 show the true and the estimated trajectories using the NNSF technique for targets 1–4, respectively. Figs. 15–18 show the same plots in case of the proposed DAPS technique. It is clear from Figs. 11–18 that the proposed DAPS technique has much better performance than the NNSF technique for all targets. In case of the NNSF, the clutter pulls the tracker away from the targets positions and poor tracking results are obtained. Figs. 19–22 show the true and the estimated trajectories using the JPDA technique for targets 1–4, respectively. Fig. 23 compares the JPDA technique and the proposed DAPS method in terms of the total measurement error for target 1. Fig. 23 shows that the tracking performance of the JPDA is slightly better than the tracking performance of the proposed DAPS technique. Similar plots and tracking results are obtained for targets 2–4. The degree of consistency between the possibility weights and the probability weights is calculated using Eq. (16) as an average of values over 600 samples and 50 runs and is found to be 0.819.

7. Conclusion

A joint possibilistic data association technique for tracking multiple targets in a cluttered environment has been proposed. Unlike the joint probabilistic data association technique, which determines joint probabilities over all targets and hits and updates the predicted target state estimate using a probability weighted sum of innovations, the proposed joint possibilistic data association technique determines joint possibilities over all targets and hits and updates the predicted target state estimate using a possibility weighted sum of innovations. The possibility weights are determined by the noise covariance matrices and the current received hits such that the total sum of the distances between all measurements and targets is minimized. Unlike many fuzzy data association techniques, which include fuzzification, fuzzy inference mechanism, fuzzy knowledge base, defuzzification, and sometimes iterative methods, the proposed association technique performs multitarget tracking based on a possibility matrix of hits to trajectories. Comparison of the required computations shows that the proposed joint possibilistic data association technique greatly reduces the computational complexity compared to the joint probabilistic data association technique. Simulated tracking systems, in the case of maneuvering and non-maneuvering targets in a cluttered environment, demonstrated the efficiency and the feasibility of the proposed association technique. It has been found that the tracking performance of the proposed association technique is much better than the tracking performance of the nearest neighbor standard filter technique and is close to the tracking performance of the joint probabilistic data association technique.

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